Module Summary: Reproducible Data Workflow with the Titanic Dataset

Author: Sandip Dey
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GitHub Repository: https://github.com/sandipdey2/ai-programming-foundations-project

Overview

This project implements a complete, reproducible data science workflow on the Titanic passenger dataset. The notebook loads the data, applies modular cleaning functions, performs exploratory analysis, generates three visualizations, and documents findings. The dataset used is the classic Titanic – Machine Learning from Disaster dataset (https://www.kaggle.com/c/titanic/data), containing information on 891 passengers. The complete project, including the notebook, requirements file, and version-controlled history, is available at https://github.com/sandipdey2/ai-programming-foundations-project.

Dataset Description

The Titanic dataset records demographic and travel information for passengers aboard the RMS Titanic, along with a binary survival indicator. After loading, the raw data contains 891 rows and 12 columns. Key variables focused on in this analysis include Survived (target), Sex, Pclass (passenger class), Age, SibSp, Parch, Fare, and Embarked. Derived features such as FamilySize, IsAlone, and Title were also created to support deeper exploration. The data is widely used in introductory data science education because it contains realistic missing values and clear subgroup differences.

Workflow Description

The workflow follows a linear, reproducible sequence:

1. **Ingestion** — The CSV is loaded with `pd.read_csv`.

2. **Cleaning** — Two functions handle missing values (median imputation for Age, mode for Embarked, drop Cabin) and create derived features.
3. **Exploratory analysis** — A dedicated function computes summary statistics, group-wise survival rates, and a correlation matrix.
4. **Visualizations** — Three plots examine survival by sex, survival rate by class, and age distribution by survival status.
5. **Summary** — Findings are interpreted in a dedicated Markdown section.

All steps are implemented as reusable functions with docstrings, supporting future extension.

Key Decisions and Assumptions

Cleaning choices prioritized simplicity and robustness. Median imputation for Age was selected because it is less sensitive to outliers than the mean (Wickham, 2014). Cabin was dropped because more than 70% of values were missing, making it unreliable for analysis. The exploratory analysis focused on Sex, Pclass, and family structure because these variables showed the strongest univariate associations with survival. Each plot was designed to answer a specific question: Figure 1 shows the raw count difference by sex, Figure 2 shows the ordered survival gradient by class, and Figure 3 reveals age patterns that are partially obscured in summary statistics. These decisions align with tidy data principles, in which each observation is a row and each variable is a column, facilitating clear visualization and analysis (Wickham, 2014).

Results and Interpretation

The analysis reveals strong patterns. Women survived at roughly three to four times the rate of men (Figure 1). First-class passengers had markedly higher survival rates than second- or third-class passengers (Figure 2). Age distributions show that children had elevated survival relative to adults in certain groups (Figure 3). Traveling alone was associated with lower survival. These findings are consistent with historical accounts of the “women and children first” protocol and class-based access to lifeboats. The patterns are descriptive rather than causal.

Responsible Practice (Bias and Data Quality)

Median imputation of Age can introduce bias if missingness is related to unobserved factors that also affect survival (for example, if poorer passengers were less likely to have ages recorded). Dropping Cabin removes potentially useful information about deck location. To reduce risk, the analysis reports the proportion of missing values before imputation and treats results as exploratory. In a production setting, multiple imputation or indicator variables for missingness would be preferable. The historical nature of the data also means modern fairness concerns (e.g., protected attributes) must be interpreted carefully.

Reproducibility

The project is designed so that another analyst can reproduce the results exactly. Dependencies are pinned in `requirements.txt`. The notebook is self-contained and uses relative paths. Version control follows a feature-branch workflow: development occurred on a `develop` branch with multiple atomic commits, then merged to `main`. The full repository, including all commits and branches, is available at https://github.com/sandipdey2/ai-programming-foundations-project. This approach mirrors the open, reproducible workflow advocated by Danchev (2022), who emphasizes the use of Jupyter notebooks, version control, and documented pipelines so that others can rerun and extend the analysis without ambiguity.

Future Integration Reflections

(1) If this data were used to train a machine learning model, the workflow would add a train/test split (or cross-validation), feature encoding (one-hot for Sex, Title, Embarked; ordinal for Pclass), and careful handling of the target to avoid leakage. Feature selection or regularization would become important.

(2) Before training a neural network, one would check for class imbalance, scale continuous features (Age, Fare, FamilySize), and verify that the imputed values do not dominate the signal. Early stopping and validation curves would be essential given the modest sample size.

(3) Agentic automation could help by generating additional diagnostic plots or flagging high-missingness columns, but it would be risky if an

agent silently dropped variables or chose imputation strategies without human review, potentially introducing undetected bias.

References

Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. *Journal of Open Source Education, 5*(56), 156. https://doi.org/10.21105/jose.00156

Wickham, H. (2014). Tidy Data. *Journal of Statistical Software, 59*(10). https://doi.org/10.18637/jss.v059.i10